

MANAS AGRAWAL

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EDUCATION

Northeastern University

Masters of Science – Computer Science | GPA: 3.83

Boston, MA

Sep 2024 – Dec 2026

- Relevant Coursework: Algorithms, System Design, Principles of Programming Languages
- Graduate Teaching Assistant for Fundamentals of Software Engineering (CS 4530)

WORK EXPERIENCE

Hydrow

Software Engineer Intern

Boston, MA

Jan 2026 – Present

- Built REST APIs using NestJS, PostgreSQL, and Redis, serving real-time athlete performance data across Hydrow's platform
- Engineered async task queue with bounded concurrency to safely batch-process movement records in parallel, preventing resource exhaustion across shared DB and Redis instances in a multi-container Kubernetes environment
- Debugged infra-level duplicate log issue by tracing the full observability pipeline — Kubernetes containers → S3 → SQS → Axiom — isolating root cause to firmware safety errors rather than application code

Studio Graphene

Software Engineer

Gurgaon, India

May 2021 – Jul 2024

- Built serverless engineering analytics platform using AWS Lambda to pinpoint bottlenecks like high PR wait times, frequent build failures, and blocked dependencies, reducing cycle time and boosting engineering velocity by 23%
- Built distributed, event-driven microservices using AWS SQS messaging queues for decoupled data ingestion and processing, reducing response latency by ~40%
- Ingested raw data from 10k+ weekly events from third party sources into Elasticsearch and ran aggregation queries to calculate high level metrics
- Built automated retry scripts to fetch failed messages from DynamoDB and reprocess via SQS batch writes, eliminating manual intervention and reducing message failure resolution time from hours to minutes
- Built scalable Node.js/TypeScript REST API backend processing 2M+ API requests daily, designed PostgreSQL schema with partitioning and indexing strategies to handle 500GB+ product catalog data and real-time cross-region inventory sync
- Deployed docker containerized microservices on AWS ECS, built CodePipeline CI/CD workflows, configured CloudWatch alarms, and integrated SES for email notifications on deployment failures and system alerts

Studio Graphene

Junior Software Engineer

Gurgaon, India

Nov 2020 – May 2021

- Developed real-time tracking and competitive scoring algorithm using Node.js/PostgreSQL with read replicas and connection pooling to serve 100K+ active users, for oddchecker's betting platform
- Engineered production-ready Python Django backend REST framework which was adopted by 10+ teams
- Implemented Redis caching, cutting repeated Algolia API calls and reducing response latency from ~300ms to ~100ms
- Implemented TDD and Trunk-based development practices using Jest and PyTest, achieving 20% faster release cycles

OPEN SOURCE PROJECTS

[NestJs Backend Boilerplate](#) | *Nodejs, TypeScript, NestJs, Docker, Git, Zod, Biome, Winston*

- A production-ready boilerplate with all the essentials like JWT authentication, RBAC, validation, logger, database, Prisma ORM, Docker and Swagger OpenAPI Specs pre-configured. It has 45 stars and 10 forks so far

[Performance Monitoring Tool](#) | *Nodejs, TypeScript, AWS X-Ray, npm, OpenTelemetry, cloudwatch, Jaeger, Prometheus*

- Built and published a package on npm, enabling end-to-end distributed tracing across APIs, DB queries, and async tasks thereby reducing mean debugging time by ~60%
- It has over 450+ downloads on npm

TECHNICAL SKILLS

Languages:

JavaScript, TypeScript, Java, C++, Python

Frameworks:

Node.js, Express.js, NestJS, Django, Laravel, React

Database:

SQL, Postgres, MySQL, NoSQL, MongoDB, Elasticsearch, Amazon DynamoDB, Firebase

Cloud & DevOps:

Cloud Computing, AWS (Lambda, S3, SQS, DLQ, X-Ray), Docker, CI/CD, cloud infrastructure

Others:

Git, Sentry, OpenTelemetry, linux, REST APIs, web architecture, frontend, agile, SDLC